Class 21

# The Useful State: Public Goods and State Capacity

*Part IV — Governing Capitalism's Consequences*

The American Economy — draft course materials (proposal under discussion)

## The point of today's class

*A successful market economy needs a capable but limited state: government is most justified when it solves specific collective-action problems that markets underprovide — defense, courts, infrastructure, public health, basic research — and its actions must be judged by costs, incentives, evidence, and institutional design.*

## Why it matters

- Would you voluntarily pay for national defense? For streetlights? For disease surveillance? If everyone benefits whether or not they pay, markets underprovide.
- Reject two bad stories:
  - **Bad story 1**: markets solve everything, so government is mostly an obstacle.
  - **Bad story 2**: if something is good, government should provide or subsidize it.
- Better: the case for government is never "this is important." It is "private markets have a *specific failure*, and government can plausibly improve the outcome more than it worsens it."
- That last clause is crucial — and it is today's discipline.

# Today

1. Public goods: rivalry and excludability
2. The free-rider problem
3. What the federal government actually does
4. Infrastructure and state capacity
5. Public health: the clean-water story
6. Knowledge spillovers; Coase, Ostrom, and civil society
7. Government failure

## The key concept: rivalry and excludability

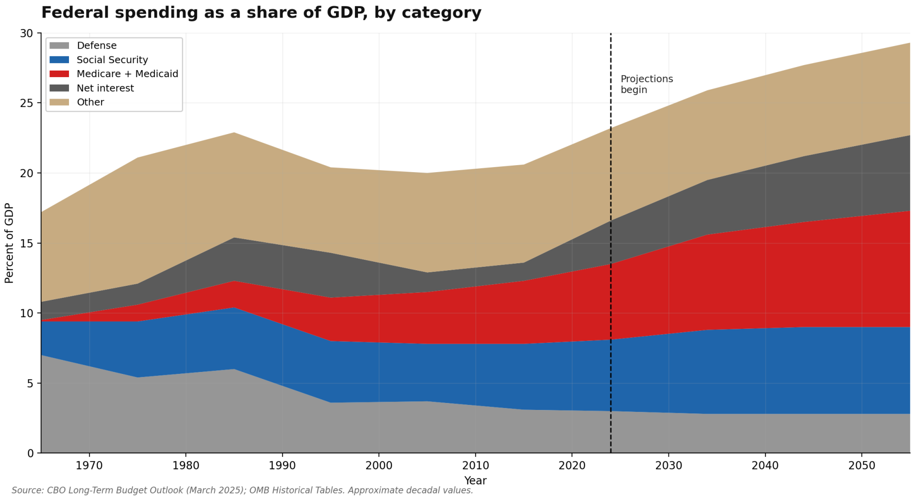
| | Excludable | Nonexcludable |
|---|---|---|
| **Rival** | Private good: sandwich, car | Commons: fishery, groundwater |
| **Nonrival** | Club good: streaming, toll road | Public good: defense, knowledge |

- **Nonrival**: my use does not reduce yours. **Nonexcludable**: hard to keep nonpayers from benefiting.
- Samuelson formalized “collective consumption goods” — the pure public good is both nonrival and nonexcludable.
- “Public good” does *not* mean “good for the public.” It names a specific incentive problem.

## The free-rider problem

- Ten roommates share a kitchen. Everyone wants it clean; nobody wants to be the one who cleans. The kitchen stays dirty.
- Scale it up:
  - roommates $\rightarrow$ national defense
  - group-project effort $\rightarrow$ basic research
  - neighborhood snow removal $\rightarrow$ roads
  - vaccination $\rightarrow$ herd immunity
  - church mutual aid $\rightarrow$ local charity
- If people can benefit without paying, voluntary private provision may be too low — even when everyone values the good.
- This is the core economic argument for taxation: compel the contribution that free riding would otherwise erode.

# Federal spending as a share of GDP, by category




*Source: Office of Management and Budget, Historical Tables; Congressional Budget Office.*

# Public goods are not government goods

| Government provides | Public good? |
|---|---|
| National defense | Close to a pure public good |
| Social Security check | Transfer / social insurance |
| Medicare payment | Social insurance, not a public good |
| Highway | Club / common good; congestion matters |
| Public school | Private benefits plus externalities |
| Farm subsidy | Transfer / political economy |

- Federal spending in FY2025: about $7.0 trillion, roughly 23% of GDP — heavily social insurance, health, defense, and interest.
- Roads, schools, police, and water are mostly state and local.
- Not a public good does not mean bad — but it needs a different justification. That is next class: taxes, insurance, and debt.

# Infrastructure creates markets

| Infrastructure | Economic role |
|---|---|
| Roads / highways | Networks, spillovers, congestion |
| Ports | Trade infrastructure |
| Water / sewer | Public-health externalities |
| Electricity grid | Natural-monopoly network |
| Courts | Enforcing contracts and property |
| Standards, GPS, weather data | Information public goods |

- Class 7 callback: railroads, highways, and broadband did not merely move goods — they *created markets*.
- Some of the best government investments do not replace markets; they make markets possible.

## State capacity matters as much as size

| Weak state | Capable state |
|---|---|
| Cannot tax reliably | Raises revenue with legitimacy |
| Cannot enforce contracts | Courts and predictable rules |
| Cannot maintain infrastructure | Builds and repairs systems |
| Corruption high | Procurement accountable |
| Projects slow and costly | Delivers public goods effectively |

- The question is not only *how much* government — it is whether government can actually deliver: tax, build, enforce, and administer.
- A capable but limited state is what makes markets possible.

# Public health: the clean-water story

- CDC's ten great public-health achievements: vaccination, motor-vehicle safety, safer workplaces, control of infectious disease, safer foods, fluoridation, and more.
- **Cutler and Miller**: clean-water technologies were responsible for nearly *half* of the total mortality reduction in major U.S. cities in the early twentieth century — and about three-quarters of the infant-mortality reduction.
- **Ferrie and Troesken**: water purification explains 30 to 50% of Chicago's mortality decline between 1850 and 1925.
- Why this is the perfect "useful state" example: high fixed costs, network infrastructure, enormous spillovers, massive mortality benefits — a return almost no private market could capture.

# Knowledge and innovation spillovers

| Problem | Why markets underprovide |
|---|---|
| Nonrival knowledge | Many can use it once discovered |
| Spillovers | Inventor cannot capture the full return |
| Long horizons | Private returns too uncertain |
| Basic science | Unknown applications |

- Class 8 callback: Vannevar Bush, NSF, NIH, DARPA, GPS, the internet, weather satellites, land-grant universities.
- Markets commercialize ideas well — but basic knowledge spills over far beyond whoever paid for it.

## Coase, Ostrom, and alternatives to government

- **Coase**: with clear property rights and low transaction costs, private bargaining can fix externalities — two neighbors and a tree. It fails when millions are affected — air pollution, disease. Coase is not "government never matters"; it is "property rights and transaction costs matter."
- **Ostrom**: communities govern commons with their own rules, monitoring, and sanctions — irrigation districts, grazing associations, fisheries. The choice is not only market or central government.
- Utah angle: irrigation, water law, religious and community mutual aid — real collective-action institutions, not nostalgia.
- **Subsidiarity**: solve problems at the lowest level capable of handling the spillover — neighborhood, city, state, nation.

## Government failure: market failure is the start, not the end

| Market failure | Government failure |
|---|---|
| Free riding | Rent-seeking |
| Externalities | Regulatory capture |
| Monopoly | Bureaucracy |
| Underprovided research | Mission creep |
| Asymmetric information | The knowledge problem |
| Infrastructure networks | Cost overruns |

- Public choice: government is made of people responding to incentives. Who pays? Who benefits? Who is organized? Who is accountable?
- A market failure is a reason to *analyze* government action — not a blank check for it.

## Discussion

Should government provide it? Classify each: national defense, a public school, a bridge, a restaurant meal, a vaccine, basic science, rural broadband, clean air, retirement income. Rival? Excludable? What exactly is the market failure — and which institution (market, government, nonprofit, family, community) handles it best?

## Read before next class

- Ostrom, “Beyond Markets and States” (Nobel lecture) — skim for the examples of commons governance.
- CDC, “Ten Great Public Health Achievements” — public health as a public-good success story.